Bao Nguyen | ECE332 Homework 4 | 05/25/20

1.) A 1-m long dipole is excited by a 1 MHz current with an amplitude of 12 A. What is the average power density radiated by the dipole at a distance of 5 km in a direction that is 45∘ from the dipole axis? (8 points)

5000 m

$$\lambda = \frac{c}{f} = \frac{3\times10^8 \text{ m/s}}{1\times10^6 \text{ Hz}} = 300 \text{ m}$$

$$\frac{l}{\lambda} = \frac{1}{300} = \frac{1}{300} \text{, which is } < \frac{1}{50} \text{ so Hertzian}$$

Avg pwr density

$$S(R,\theta) = \frac{\eta_0 k^2 I_0^2 l^2}{32\pi^2 R^2}\sin^2(\theta)$$

$$= \frac{120\pi\left(\frac{\pi}{150}\right)^2 (12)^2 (1)^2}{32\pi^2\, 5000^2}\sin^2(45^\circ)$$

nW/m^2

$$\boxed{S(5000, 45^\circ) = 1.51\ \cancel{pW/m^2}}$$

Given: $\eta_0 = 120\pi$

$K = \frac{2\pi}{\lambda} = \frac{2\pi}{300} = \frac{\pi}{150}$

$\lambda = 300$

$I_0 = 12$ A

$l = 1$m

$R = 5000$m

$\theta = 45^\circ = \frac{\pi}{4}$ ~~rads~~

2.) The normalized radiation intensity of a certain antenna is given by

$$F(\theta) = \exp(-20\theta^2) \text{ for } 0 \leq \theta \leq \pi,$$

where θ is in radians. Determine: (15 points total)

a. the half power beamwidth, (5 points) (solve for β) (rads mode)

$F(\theta_{1/2}) = 0.5$

$e^{-20\theta_{1/2}^2} = 0.5$

$-20\theta_{1/2}^2 = \ln(0.5)$

$\theta_{1/2} = \pm 0.1862$ rads

$\beta = 2\theta_{1/2} = 2(0.1862) = \boxed{0.3723 \text{ rads}}$

b. the pattern solid angle, (5 points) Ω_p

$\Omega_p = \int_0^{2\pi} \int_0^{\pi} F(\theta)\sin(\theta)\, d\theta\, d\phi$

$= 2\pi \int_0^{\pi} e^{-20\theta^2} \sin(\theta)\, d\theta$

use TI-36x Pro

Literally plug it in as is but use X instead of θ

$\boxed{\Omega_p = 0.156 \text{ sr}}$

$(155.7771574 \times 10^{-3})$

c. the antenna directivity. (5 points)

$D = \frac{4\pi}{\Omega_p} = \frac{4\pi}{0.156} = \boxed{80.67 \text{ (unit less)}}$

3.) For a dipole antenna of length l = λ/4, (15 points total)

a. determine the directions of maximum radiation, (5 points)

$$\lim_{\theta\to 0} \frac{f(\theta)}{g(\theta)} = \lim_{\theta \to 0} \frac{f'(\theta)}{g'(\theta)}$$

Base Formula

$$S(\theta) = \frac{15 I_0^2}{\pi R^2}\left[\frac{\cos\left(\frac{\pi l}{\lambda}\cos\theta\right) - \cos\left(\frac{\pi l}{\lambda}\right)}{\sin\theta}\right]^2$$

$$l = \frac{\lambda}{4} \qquad \frac{\pi l}{\lambda} = \frac{\pi}{\lambda}\cdot\frac{\lambda}{4} = \frac{\pi}{4}$$

$$= \frac{15 I_0^2}{\pi R^2}\left[\frac{\cos\left(\frac{\pi}{4}\cos\theta\right) - \cos\left(\frac{\pi}{4}\right)}{\sin(\theta)}\right]^2$$

$$= \underbrace{\frac{15 I_0^2}{\pi R^2}}_{\text{constant no } \theta \text{ so let } = S_0}\left[\frac{\cos\left(\frac{\pi}{4}\cos\theta\right) - \frac{\sqrt{2}}{2}}{\sin(\theta)}\right]^2$$

$$\frac{S(\theta)}{S_0} = S_0\left[\frac{\cos\left(\frac{\pi}{4}\cos\theta\right) - \frac{\sqrt{2}}{2}}{\sin(\theta)}\right]^2$$

Use ti-36x pro

$$\frac{d}{dx}\left(\cos\left(\frac{\pi}{4}\cos(x)\right) - \frac{\sqrt{2}}{2}\right)\Big|_{x=x} = 0$$

$$\frac{d}{dx}(\sin\theta)\Big|_{x=x} = 999.9\ldots\times 10^{-3} = 1$$

$$S(\theta) = S_0\left[\frac{0}{1}\right]^2$$

$S(\theta) = 0$ (peak is somewhere between $0, \pi$)

use calc

θ	30°	60°	90°
S/S_0	0.020	0.063	0.0860

$$\boxed{\theta = 90^\circ} = \frac{\pi}{2}$$

b. obtain an expression for Smax, (5 points)

$S_{max} = S(\theta)$

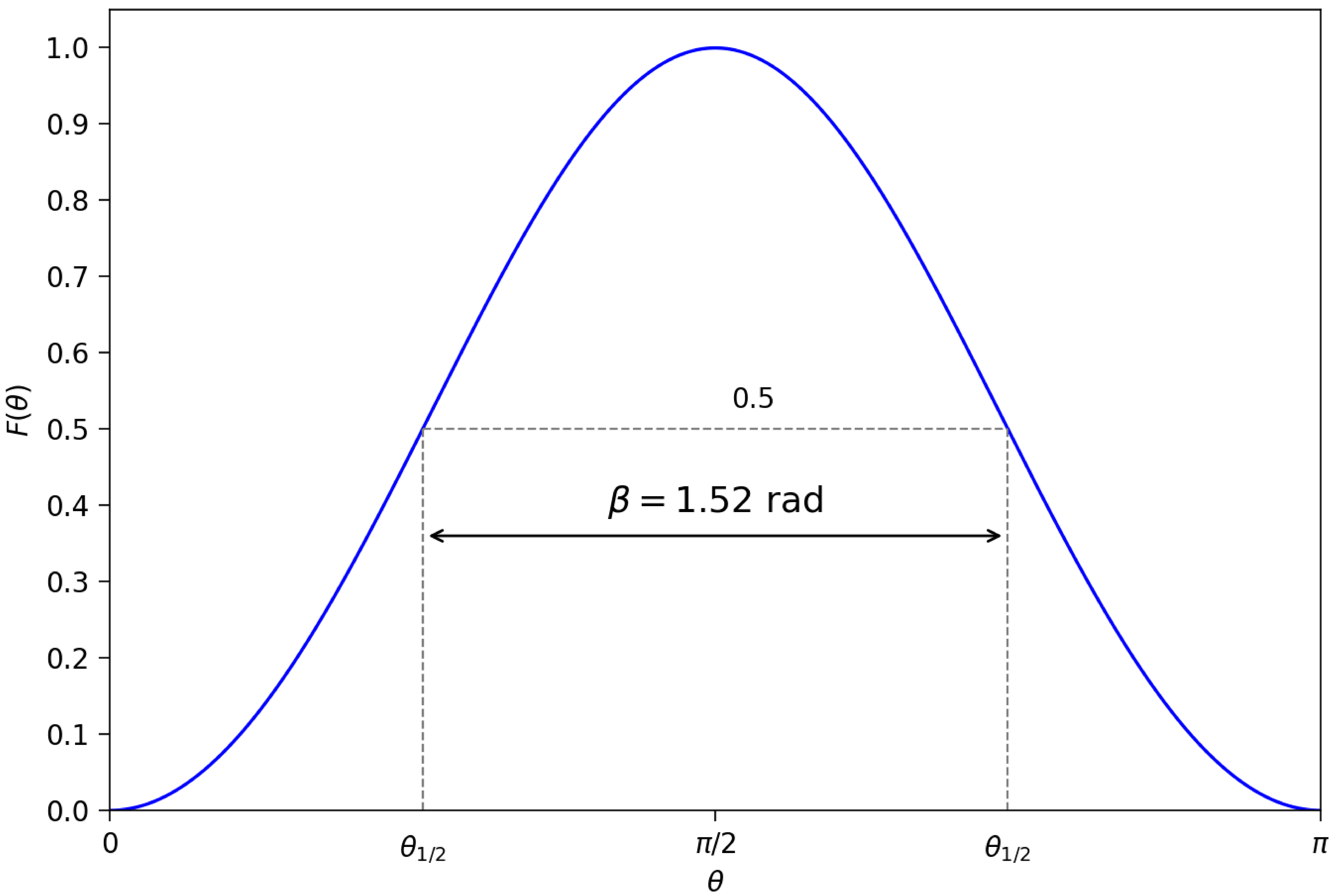
$$S\left(\frac{\pi}{2}\right) = \frac{15 I_0^2}{\pi R^2}\left[\frac{\cos\left(\frac{\pi}{4}\cos\left(\frac{\pi}{2}\right)\right) - \frac{\sqrt{2}}{2}}{\sin\left(\frac{\pi}{2}\right)}\right]^2$$

$$= 0.0858\,\frac{15 I_0^2}{\pi R^2}$$

c. generate a plot of the normalized radiation pattern F (θ). (5 points)

$$F(\theta) = \frac{S(\theta)}{S_{max}} = \frac{\frac{15 \cancel{I_0^2}}{\cancel{\pi} R^2}\left[\frac{\cos\left(\frac{\pi}{4}\cos(\theta)\right) - \frac{\sqrt{2}}{2}}{\sin(\theta)}\right]^2}{\frac{15 \cancel{I_0^2}}{\cancel{\pi} R^2}\; 0.0858}$$

$$= \boxed{11.66\left[\frac{\cos\left(\frac{\pi}{4}\cos(\theta)\right) - \frac{\sqrt{2}}{2}}{\sin(\theta)}\right]}$$



4.) Determine the effective area of a half-wave dipole antenna at 100 MHz, and compare it with its physical cross section if the wire diameter is 2 cm. (5 points)

Wave length $\lambda = \frac{c}{f} = \frac{3 \times 10^8 \ m/s}{100 \times 10^6 Hz} = 3m$

Since half wave diepole $D = 1.64$ from cheat sheed

Effective Area

$A_e = \frac{\lambda^2 D}{4\pi} \ m^2$

$= \frac{(3^2)(1.64)}{4\pi}$

$\boxed{A_e = 1.17 m^2}$

Physical cross section

l for half wave dipole is $l = \frac{\lambda}{2}$, $d = 0.02m$

$A_p = l \cdot d = \frac{\lambda}{2} d = \frac{3}{2}(0.02) = \boxed{0.03 \ m^2}$

Ratio

$\frac{A_e}{A_p} = \frac{1.17}{0.03} = \boxed{39.2}$

The effective area is 39 x larger than physical cross section.

5.) A half-wave dipole TV broadcast antenna transmits 1 kW at 50 MHz. What is the power received by a home television antenna with a 3 dB gain if located at a distance of 30 km? (5 points)

$$P_r = G_t G_r \left(\frac{\lambda}{4\pi R}\right)^2 P_t$$

$$\lambda = \frac{c}{f} = \frac{3\times10^8}{50\times10^6} = 6m$$

Gain transmitting

$G_t = D$ for lossless so from chart sheet for half wave dipole

$D = 1.64 = G_t$

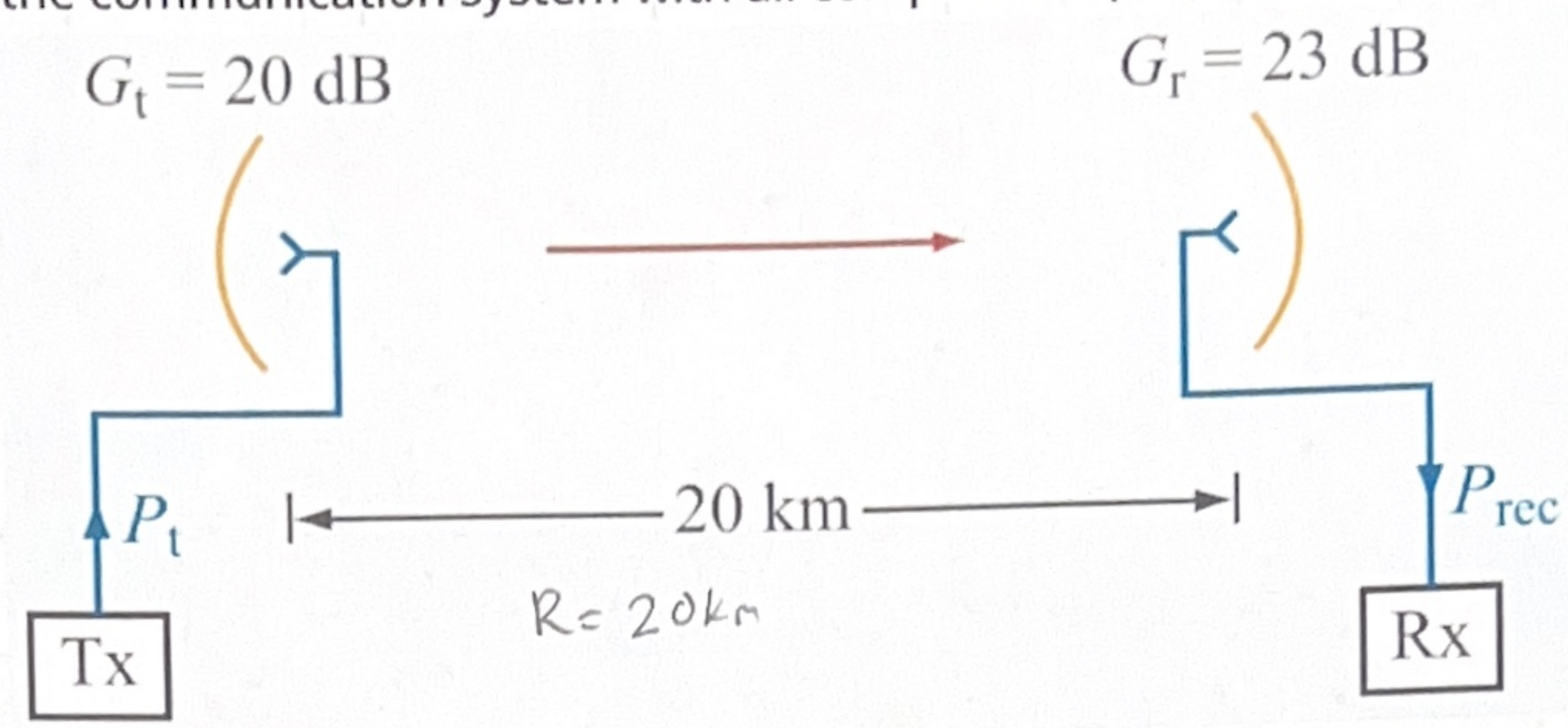
Gain receiving

$G_r = 3dB$ (Given) $\rightarrow$ Convert to linear = G_r linear $= 10^{3db/10} \approx 2$

$R = 30km = 30\times10^3 m$ $\qquad$ $P_t = 1kW = 1000W$

$$P_r = (1.64)(2)\left(\frac{6}{4\pi(30\times10^3)}\right)^2(1000) = \boxed{8.31\ nW}$$

6. Consider the communication system with all components properly matched shown below.



Determine the following if P_t = 10 W and f = 6 GHz.

(a) What is the power density at the receiving antenna (assuming proper alignment of antennas)?

$S_r = \frac{G_t P_t}{4\pi R^2} = \frac{(100)(10)}{4\pi(20\times10^3)^2} = \boxed{198.9\ nW/m^2}$

$G_{t\,linear} = 10^{20/10} = 100$

(b) What is the received power?

$P_r = G_t G_r \left(\frac{\lambda}{4\pi R}\right)^2 P_t$

$= (100)(199.5)\left(\frac{0.05}{4\pi(20\times10^3)}\right)^2 10$

$= 7.8969\times10^{-9}$

$\boxed{P_r = 7.90\ nW}$

$G_r = 10^{23/10} = 199.5$

$\lambda = \frac{c}{f} = \frac{3\times10^8}{6\times10^9} = 50\times10^{-3}\,m = 0.05$

(c) If T_sys = 1000 K and the receiver bandwidth is 20 MHz, what is the signal-to-noise ratio in decibels?

$SNR_{dB} = 10\log_{10}\frac{P_r}{P_N}$

$= 10\log_{10}\frac{7.90nW}{276\times10^{-15}}$

$= \boxed{44.6\ dB}$

$P_N = k_B T_{sys} B$

$= (1.38\times10^{-23})(1000)(20\times10^6)$

$= 276\times10^{-15}$ Wb

7. An antenna with a circular aperture has a circular beam with a beamwidth of 3° at 20 GHz.

(a) What is the antenna directivity in dB?

$\beta = 3° \cdot \frac{\pi}{180} = 0.05236$ rads

$D_{lin} = \frac{4\pi}{\beta^2} = \frac{4\pi}{(0.05236)^2} = 4584$ unitless

$D_{dB} = 10 \log(4584) = \boxed{36.6 \text{ dB}}$

(b) If the antenna area is doubled, what will be the new directivity and new beamwidth?

$D_{new} = 2D_{old} = 2(4584) = 9168$ unitless

$D_{new\,db} = 10 \log_{10}(9168) = \boxed{39.6 \text{ db}}$

$B_{new} = \frac{B_{old}}{\sqrt{2}} = \frac{0.05236 \text{ rads}}{\sqrt{2}} = 0.037$ rad

$B_{new}° = 0.037 \times \frac{180}{\pi} = \boxed{2.12°}$

(c) If the aperture is kept the same as in (a), but the frequency is doubled to 40 GHz, what will the directivity and beamwidth become then?

$f_{new} = 40 \times 10^9$ Hz

D_{new} $4D_{old} = 4(4584) = 18336$

$D_{new\,db} = 10 \log(D_{new}) = \boxed{42.63 \text{ db}}$

$B_{new}° = \frac{B_{old}}{2} = \frac{3°}{2} = \boxed{1.5°}$

8. Find and plot the normalized array factor and determine the half-power beamwidth for a 5-element linear array excited with equal phase and a uniform amplitude distribution. The interelement spacing is 3λ/4.

$N=5$ $\quad k=\frac{2\pi}{\lambda}$ $\quad d=\frac{3\lambda}{4}$ $\quad \gamma = kd\cos\theta$ $\quad d=\frac{3\lambda}{4}$

$$F_a(\theta) = \frac{\sin^2(N\gamma/2)}{\sin^2(\gamma/2)}$$

$$\gamma = \frac{2\pi}{\lambda}\cdot\frac{3\lambda}{4}\cos(\theta) = \frac{3\pi}{2}\cos\theta$$

$$= \frac{\sin^2\left(\frac{5\left(\frac{3\pi}{2}\right)\cos\theta}{2}\right)}{\sin^2\left(\frac{\frac{3\pi}{2}\cos\theta}{2}\right)}$$

$$= \frac{\sin^2\left(\frac{15\pi}{4}\cos\theta\right)}{\sin^2\left(\frac{3\pi}{4}\cos\theta\right)}$$

$$F_{a}^{norm} = \frac{F_a}{N^2} =$$

$N^2 = 5^2 = 25$

$$\boxed{F_a^{norm} = \frac{1}{25}\,\frac{\sin^2\left(\frac{15\pi}{4}\cos\theta\right)}{\sin^2\left(\frac{3\pi}{4}\cos\theta\right)}}$$

Uniform Broadside array

$$B = 0.886\frac{\lambda}{Nd}\ (\text{rad})$$

$$Nd = 5\cdot\frac{3\lambda}{4} = \frac{15\lambda}{4}$$

$$B = 0.886\cdot\frac{\lambda}{\frac{15\lambda}{4}}$$

$$= 0.886\cdot\cancel{\lambda}\cdot\frac{4}{15\cancel{\lambda}}$$

$$= 0.2363\ \text{rad}$$

$$B_{deg} = B\left(\frac{180}{\pi}\right)$$

$$= 0.2363\left(\frac{180}{\pi}\right)$$

$$= \boxed{13.5^\circ}$$